

Genetic Information: Cell to Gene

I can describe where genetic information is stored and how chromosomes, DNA, genes and the genome are related.

LOCATE

cell to gene

RELATE

part and whole

LIMIT

models are not to scale

[Open the original interactive lesson](#)

What success looks like

- I can order the genetic-information hierarchy.
- I can define chromosome, DNA, gene and genome.
- I can correct a reversed or scale-distorted hierarchy.

Retrieve and reconnect

- State where chromosomes are found in a eukaryotic cell.
- Recall why daughter cells receive identical chromosome sets in mitosis.
- Distinguish magnification from resolution.
- Lead-in: predict whether a gene is larger or smaller than a chromosome.

Which order moves from a large biological structure to a smaller part of its genetic material?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Cell → nucleus → chromosome → DNA → gene
- B. Gene → cell → DNA → nucleus → chromosome
- C. Cell → protein → nucleus → gene → chromosome

Look closely · what is the evidence?

CELL · contains nucleus

CHROMOSOME · one long DNA molecule with proteins

GENE · section of DNA

Look closely · what is the evidence?



Cell and nucleus



To DNA



To gene

Words we will use

- chromosome — packaged DNA
- gene — section of DNA
- genome — all an organism's DNA

Watch the model

- Start with a eukaryotic body cell and locate its nucleus.
- Inside the nucleus, identify chromosomes; each chromosome contains one long DNA molecule associated with proteins.
- Zoom conceptually to DNA: a gene is a section of that DNA, and the genome is all the organism's DNA.

Which relationship is correct?

- A. A gene is a section of DNA located on a chromosome.
- B. A chromosome is a section of a gene located outside the cell.
- C. A genome is one protein made by one gene.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- A gene's base sequence provides information for making a particular protein; the lesson does not require Higher-tier protein-synthesis stages.
- Different genes occupy different sections of chromosome DNA, while the genome includes all chromosomes and other DNA in the organism.
- Chromosomes are not permanently X-shaped; that familiar shape represents a condensed, copied chromosome during cell division.

Find and repair the misconception

A gene contains DNA, and DNA contains chromosomes.

A. Keep it: the smallest item contains the larger ones.

B. Repair it: chromosomes contain DNA; genes are sections of DNA.

C. Repair it: genes are separate cells inside the nucleus.

Genetic hierarchy placement lab

Place each description at the correct level, then read the complete hierarchy aloud or by pointing.

- CELL
- NUCLEUS
- CHROMOSOME
- DNA
- GENE

Genetic hierarchy placement lab

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- CELL / NUCLEUS LEVEL
- CHROMOSOME / DNA LEVEL
- GENE / GENOME IDEA

Use contains and section of to describe the full hierarchy without reversing it.

Your independent mission

Build and annotate the genetic-information hierarchy, then explain one model strength and one limitation.

Record: An ordered hierarchy, accurate definitions and one honest statement about scale or shape.

Choose your support and challenge

- Supported: Order five icon-backed cards and complete: a gene is a section of ____.
- Standard: Build the full hierarchy and write two linked relationship sentences.
- Stretch: Explain the hierarchy, include genome and critique the permanent-X and scale misconceptions.

Show what you know

Describe how a chromosome, DNA and a gene are related.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.